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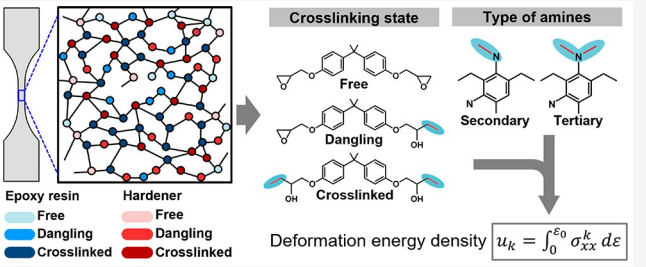
Subcontinuum Interpretation of Mechanical Behavior for Cross-Linked Epoxy Networks

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ABSTRACT: The origin of the mechanical properties of highly cross-linked epoxy networks was theoretically investigated from a subcontinuum perspective. By use of all-atom molecular dynamics (MD) simulations, the macromolecular network of epoxy formed during the cross-linking reactions was classified into subgroups according to their bonding relationship. The deformation energy density applied to the entire system under mechanical loading is expressed by the contribution of each subgroup. The load transfer capabilities according to the chemical bonding state between resin and hardener were then quantified at the atomic level. On the basis of the results, an analytic blending model was established that can predict mechanical properties of the cross-linked epoxy according to its chemical composition and associated network topology. It was confirmed that the proposed model successfully predicts the mechanical properties of materials for the range of composition ratios that can be considered in actual synthesis as well as an in-depth analysis of the individual molecular components.



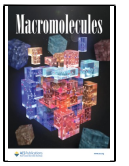
1. INTRODUCTION

Epoxy thermosets have a wide range of applications in a variety of engineering fields, such as electronics, automotive, and aerospace.[1−3] Because of its excellent thermomechanical performance and ease of processing, epoxy is a common matrix material used as advanced composites with fibers and particles or as independent functional materials.[4−7] The most promising feature of epoxy thermosets is its high stiffness, which is imparted by the three-dimensional network structure formed during the cross-linking reaction. Because the properties of the finished products are significantly dependent on the microscopic structure of the network, it is important to design well-organized networks at the molecular scale to obtain the desired properties.[8−14]

The topological structure of the epoxy network is determined by the bonding relationships between the molecular components, which affects the overall mechanical behavior of the epoxy thermoset. The states of the components can be classified as free, dangling, and cross-linked chain, and they are interwoven together in the network structure.[15−18] Such topological features are all exhibited at the sub-10 nm scale.[19] In this regard, molecular dynamics (MD) simulations can be an effective tool to bridge the correlation between the microscopic structure of epoxy thermosets and its thermomechanical properties. Li and Strachan pioneered the evolution of the network structure during the cross-linking reaction *in silico*.[20,21] The changes in the states of the components during cross-linking were quantified, and a fundamental relationship with the mechanical properties was discovered. Estridge classified the bonding state of epoxy components based on the degree of nodes.[22] The influence of the kinetic weight factor on the development of the network morphology of epoxy during the cross-linking reaction was investigated. Although the above-mentioned studies have elucidated the topological characterization of the polymer morphology, a detailed description of the mechanical role of each molecular component in the entire macroscopic system remains unresolved. Specifically, there was no attempt regarding subcontinuum analysis to identify the effective properties of each molecule under mechanical loading, which is essential for understanding the principle of change in mechanical properties depending on the network topology.

Motivated by this need, in this study, we hierarchically analyzed the mechanical behavior of highly cross-linked epoxy networks, taking into account both chemical bond states and atomic structures of molecules. A highly cross-linked epoxy microstructure composed of the bifunctional resin and hardener was prepared in an all-atom MD environment. In addition to the elastic−plastic behavior showing the overall network structure, the effective deformation energy occupied by the resin and hardener was separately analyzed. The classification of deformation energy density according to the

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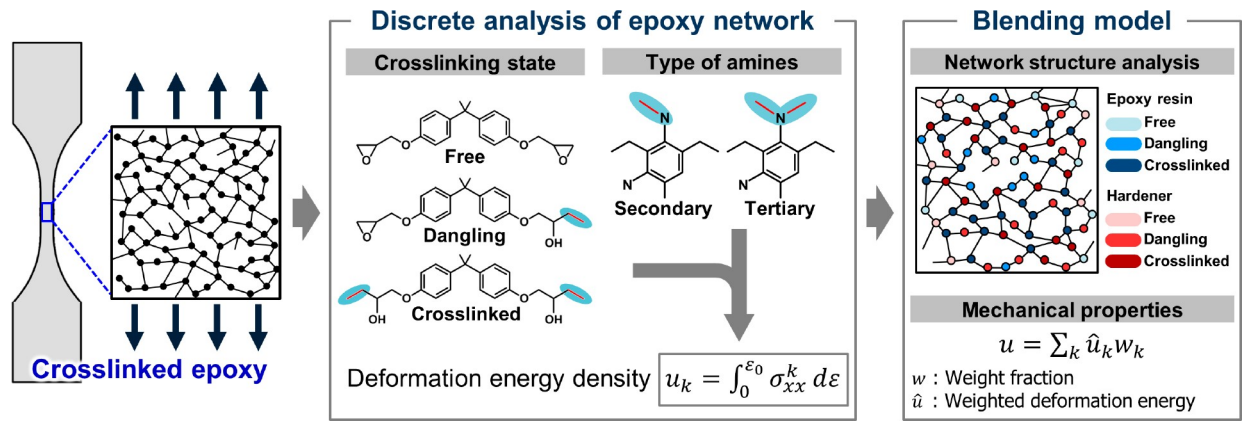


Figure 1. Scheme of the methodology developed in this study.

state of chemical bonds was followed, and finally, how the total deformation energy was distributed to all components constituting the epoxy network was identified. On the basis of the analyzed data, a rule-of-mixture-based blending model was proposed that predicts the mechanical behavior of an epoxy thermoset according to molecular composition. This makes it possible to identify the load transfer capability of each element constituting the network structure systematically. The associated underlying mechanism by which subcontinuum structures lead to the expression of macroscopic properties was also thoroughly discussed.

2. METHODOLOGY

The methodology established is described in Figure 1. During the cross-linking reaction simulation, the constituent molecules of the epoxy network are highly coupled with each other, forming a single macro-network across the periodic boundaries of the unit cell. A more precise description of this structure from the perspective of a subcontinuum reveals that molecular components with various chemical bonding states are woven together in a network. The macroscopic properties of a material can be determined as a comprehensive representation of the behavior of these individual components.

The main idea of this study is based on the assumption that the relationship between the chemical bonding structures and material properties can be numerically quantified by analyzing the responses of individual components down to the atomic scale. In this regard, it is necessary to consider the fact that even the same type of molecule has different contributions to the physical properties of the material depending on the state of the chemical bond. Therefore, the identical molecules with different cross-linking states should be considered as different components. From this point of view, network members can be classified into three categories according to the cross-linking site: free, dangling, and cross-linked chains. For the epoxy hardener, further classification according to amine type was also applied.

As a first step of this framework, this study focuses on the distribution of deformation energy due to mechanical load, which is one of the most important and fundamental properties of epoxy networks. In all-atom MD simulations, the virial stress of the modeled unit cell can be calculated by eq 1:[23]

$$\boldsymbol{\sigma} = \frac{1}{V}\left(-\sum_{i}^{N} m_i(\mathbf{v}_i\mathbf{v}_i^{\mathrm{T}}) + \frac{1}{2}\sum_{i}^{N}\sum_{j\neq i}^{N} \mathbf{r}_{ij}\mathbf{F}_{ij}\right) \quad (1)$$

where V is the volume occupied by the system, N is the number of atoms, m_i and $\mathbf{v}_i$ are the mass and velocity of the ith atom, and $\mathbf{r}_{ij}$ and $\mathbf{F}_{ij}$ are the relative position and force between the ith and jth atoms in the network, respectively. The virial stress determines the load applied to the atom from the velocity by thermal fluctuation and interatomic force. If V is set as the volume of the entire system, it is self-evident that the stress on the entire network will be equal to the linear sum of the stresses assigned to the individual members constituting the network. In particular, it has been demonstrated that the virial stress has the same physical meaning as the continuum Cauchy stress when small deformation is applied to the model in a state without the tilting of the normal stress after deformation.[24] Meanwhile, during mechanical load simulations, the increment of atomic stress can be computed individually for all atoms in the unit cell or for specific groups defined by the user. These two features of eq 1 are the key basis allowing the properties of network characterized from the subcontinuum perspective to be bridged with the macroscopic properties. Note that the process of calculating the stresses of atoms belonging to a specific region in the model used for MD simulation through the virial theorem is not a new attempt. The calculation of the critical stress for crack opening,[25−27] load transfer in the interfacial region of nanocomposites,[28,29] and stress concentration around nanodefects[30] have already been reported. However, what makes the approach proposed in this study novel is that the positions of the atoms participating in the calculation are not concentrated in one place but are evenly distributed throughout the unit cell. In other words, the previously known concept of calculating virial stress in a specific region has been extended to novel methodologies to characterize specific types of molecules that form unspecified shapes.

When the total uniaxial strain applied to the unit cell is ε_0, the deformation energy density experienced by the atomic constituents in the loading direction can be simply calculated from eq 2

$$u = \int_0^{\varepsilon_0} \sigma_{xx} \, d\varepsilon \quad (2)$$

where σ_{xx} indicates the virial stress component along the axial loading direction.

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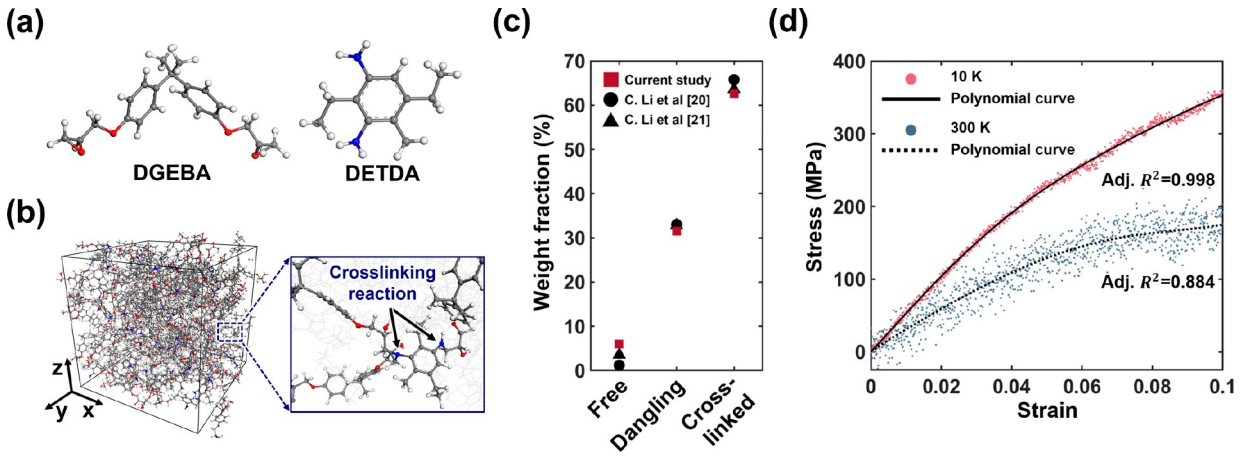


Figure 2. Cross-linked epoxy microstructure prepared in this study. (a) Molecular structure of the epoxy resin (DGEBA) and hardener (DETDA). (b) Configuration of the epoxy unit cell after the cross-linking reaction. (c) Weight fraction of free, dangling, and cross-linked chains contained in the epoxy network. (d) Stress–strain curve obtained from uniaxial tensile test simulations at different temperature conditions. Adjusted R^2 values for each regression are also presented in the figure.

To quantitatively understand the independent response of the network members under deformation, the effective energy density of the kth member (u_k) is defined. Here, the effective strain of the kth member relative to the total strain of the unit cell cannot be quantified. However, the area of interest in of this study is how energy density of the entire system distributes to each member under a given total strain, and thus information about individual strain fields is not required. Therefore, the u_k is derived from the total strain of the entire system as follows:

$$u_k = \int_0^{\varepsilon_0} \sigma_{xx}^k \, \mathrm{d}\varepsilon \tag{3}$$

where σ_{xx}^k is the virial stress component of the kth member in the axial loading direction. The deformation energy density of the entire network is now equal to the linear sum of the energy distributed to the network members as follows:

$$u = \sum_k u_k \tag{4}$$

For the independent characterization of molecules formed by specific groups of atoms, a blending model based on mixing rules was considered as follows:

$$u = \sum_k \hat{u}_k w_k \tag{5}$$

where $\hat{u}$ and w are the weighted deformation energy density and weight fraction, respectively. Because of the correspondence between the component boundaries and the regions partitioning the effects of interatomic forces, the weight deformation energy directly represents the effective properties of individual units homogeneously distributed in the epoxy network. The details are provided in the "Considerations on the Method" section of the Supporting Information. From eq 5, it is possible to quantify the change in the contribution of the epoxy network constituent members according to the composition of the resin and hardener used to synthesize the epoxy blend under real working conditions.

Before the detailed analysis process and results are discussed, the conceptual limitations of the proposed model are clarified below. First, the model assumes that molecular-scale components form spatially homogeneous phases anywhere in the network microstructure. This comes from the implication that all resin and hardener molecules that comprise the epoxy network have similar densities. Accordingly, the load-carrying capacity of the individual components was classified solely by the chemical bonding relationship, without considering the density distribution of each raw material. Second, the current model is suitable for a pristine phase without nanopores, crack defects, or inclusions. The one-to-one relationship between the weight distribution of each local spatial region and the volume fraction occupied by the components is a necessary condition for linear classification of mechanical deformation energy. Therefore, the rapid growth of the free volume or the semicrystalline behavior of the polymer matrix, which can occur during extremely large deformation processes, is not addressed in this model. Lastly, the current model does not take into account changes in the chemical bonding states during mechanical loading. Therefore, it is inappropriate to apply this model to the evaluation of environments in which covalent bonds change dynamically. Despite these limitations, the proposed model provides important insight regarding the elastoplastic behavior of a homogeneous matrix in which the bonding relationship is maintained, which will be described in detail in section 4.

3. MODELING OF CROSS-LINKED EPOXY MICROSTRUCTURES

A highly cross-linked epoxy microstructure was modeled by using all-atom MD simulations. Figure 2a shows the molecular configurations of the epoxy resin (bisphenol A diglycidyl ether, DGEBA) and hardener (diethyltoluenediamine, DETDA) used in this study. The resin and hardener molecules were randomly disposed in a periodic cubic cell at a molecular ratio of 2:1 (100 and 50 molecules, respectively). The initial density of the unit cell was set to 0.2 g/cm^3 to avoid undesired interference between molecules. Subsequently, the length of the unit cell was decreased equally in all directions until the density of the entire system reached 1.0 g/cm^3. For each incremental increase in density of 0.2 g/cm^3, the conjugate gradient method was performed with an energy convergence cutoff of 10 kcal/(mol Å) to optimize the geometry of all molecules. The unit cell was equilibrated by using the NVT

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Table 1. Structural and Mechanical Properties of Cross-Linked Epoxy Compared with Previous Studies

	approach	resin/hardener	cross-linking ratio (%)	density (g/cm^3)	elastic modulus (GPa)	
					300 K	10 K
current study	MD simulation		75.5	1.08	2.95	5.19
Qi et al.[39]	experiment				2.71	
Rafiee et al.[40]	experiment	DGEBA/DETDA		1.08	2.95	
Clancy et al.[41]	MD simulation		79.6	1.08	2.79	
Shenogina et al.[42]	MD simulation		79.9		3.21	

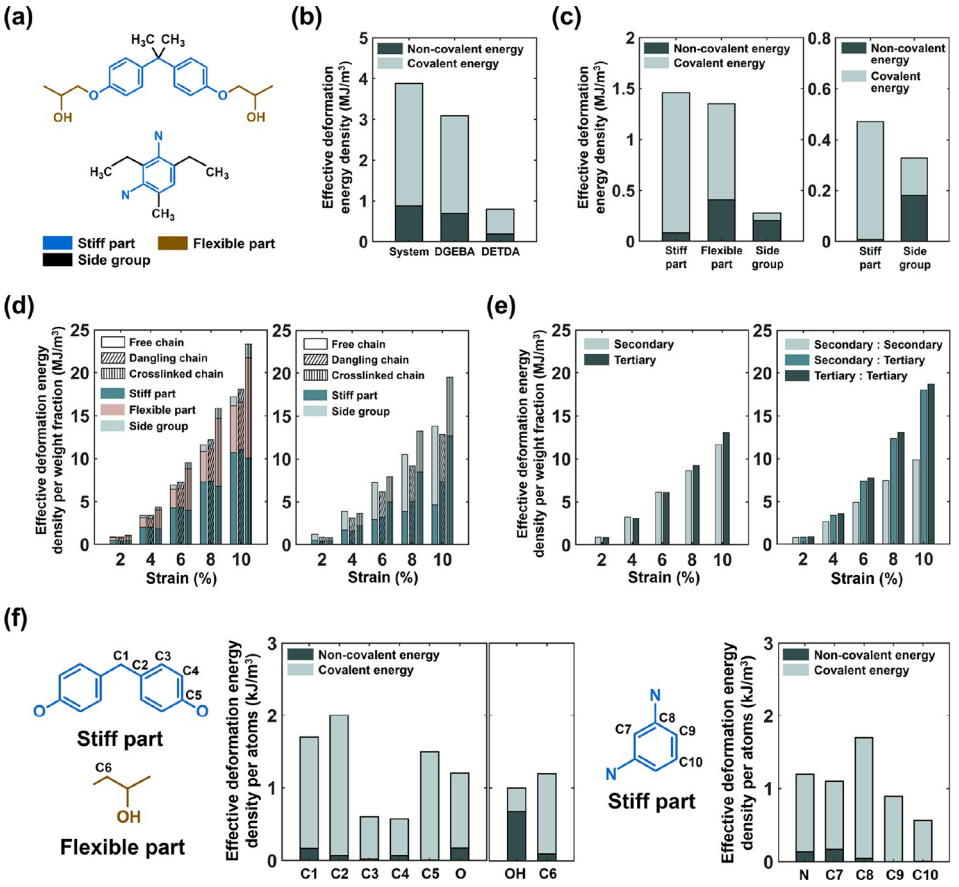


Figure 3. Discrete analysis of the mechanical response of constituents under tensile deformation. (a) Structural classification of DGEBA and DETDA. (b) Deformation energy density distributions of the entire system and each molecular component at a strain of 4%. (c) Deformation energy density distributions of the subconstituents comprising the DGEBA resin (left) and DETDA hardener (right) at a strain of 4%. (d) Weighted deformation energy density of the DGEBA resin (left) and DETDA hardener (right). The data obtained were evaluated according to the cross-linking state of each molecule and the parts constituting the molecule. (e) Classification of the energy density of the DETDA hardener according to the number of cross-linked points. The results for the dangling DETDA (left) and cross-linked DETDA (right) are displayed separately. (f) Effective deformation energy assigned to constituent atoms of the epoxy resin and hardener at a strain of 4%.

ensemble at 300 K for 1 ns, followed by the NPT ensemble at 300 K and 1 atm for 1 ns, with a time step of 1 fs.

The cross-linking reaction between DGEBA and DETDA was performed to obtain a cross-linked epoxy network structure. The reactive sites of DGEBA and DETDA in a given unit cell were defined, and the relative distance between these sites was continuously monitored during the MD run. If the relative distance was closer than the cutoff value, covalent bonds between the reactive sites are created. The two tertiary amines in a single DETDA hardener can be coupled with up to four epoxide groups. The cutoff value was initially set to 3.0 Å and gradually increased, in increments of 0.5 Å, until 75% of the epoxide groups in DGEBA reacted. The cross-linked epoxy model was equilibrated again by using the NVT ensemble at 300 K for 1 ns and NPT ensemble at 300 K and 1 atm for 2 ns. The configuration of the equilibrated epoxy microstructure is depicted in Figure 2b.

The validity of the epoxy model was secured through investigation of the network topology and mechanical tensile tests. Figure 2c shows the weight fraction by structural state of the polymer chains constituting the epoxy network. The weight fraction of the cross-linked chain was the largest (62.5 wt %), followed by dangling (31.5 wt %) and free chains (5.7 wt %). Most molecules in the network of high cross-linking density consist of cross-linked chains, which is plausible given the short chain length of the epoxy components. The magnitude of the weight fraction of each state also showed good agreement with the values reported in the literature for other epoxy networks with around 80% cross-linking ratio.[20,21]

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The mechanical properties of the cross-linked epoxy model were evaluated via uniaxial tensile tests. The modeled unit cell was elongated at a strain rate of 10^8 /s under the NPT ensemble at 1 atm until the applied strain reached 10%. Note that the strain rate conditions used in this study are typical for classical MD simulations.[31−33] As Rottler and Robbins pointed out in their study, the rate is sufficient to allow the evolved stresses to equilibrate the model of the glassy polymer material.[34] In other words, in all cases covered in this study, the deformation according to the load of the material proceeds quasi-statically. The test was performed at 300 and 10 K, and the values were averaged after obtaining the results for the *x*-, *y*-, and *z*-directions. The overall stress−strain profiles were fitted with a fifth-order polynomial regression,[35,36] and the elastic modulus was calculated from the slope of the curve at strain below 2%. The stress−strain curves of the model and its elastic modulus are presented in Figure 2d and Table 1, respectively. The model exhibited the typical behavior of a thermoset polymer (i.e., an elastoplastic response without strain softening[37,38]) until the deformation reached 10%. These results suggest that the cross-linked networks within the epoxy microstructure were sufficiently distributed throughout the periodic box boundaries and formed a highly condensed structure without localization. In particular, the elastic modulus estimated under room-temperature conditions was quantitatively consistent with other experimental and theoretical results,[39−42] thereby confirming the validity of the model prepared.

It is noteworthy that all the analysis procedures discussed hereafter were based on conditions in which temperature effects were excluded. Although the simulated results obtained at room temperature fit the experimental macroscopic characteristics very well, it is inappropriate to interpret the virial stress of each component constituting the system from a mechanical perspective. If the target temperature of a finite size is set, the velocity of the constituent atoms is forcibly controlled by the thermostat applied in the NPT ensemble during uniaxial deformation.[43,44] The kinetic energy caused by the velocity change of atoms induces harmonic oscillations of covalent bonds, and the noncovalent bonds, which are relatively weaker, function as dampers to absorb them. Because the purpose of this study is to quantify the static load transfer characteristics between components within the microstructure, there should be no distortion in the internal stress calculation due to the thermostat. Therefore, the mechanical behavior of the model was based on a tensile simulation performed at a cryogenic temperature of 10 K instead of room temperature. As shown in Figure S2, the tensile simulation at room temperature has a form in which the dynamic distortion caused by covalent stress is corrected by the attraction of noncovalent stress. Conversely, in the cryogenic tensile simulation, both covalent and noncovalent stresses increase linearly, allowing for quasi-static interpretation throughout the entire deformation process. With this as a beginning, the next section deals with the quantitative analysis of the independent properties of molecules and their bonding structures that directly affect the mechanical properties of the entire system.

All-atom modeling of the cross-linked epoxy was performed by using the commercial MD software, Materials Studio 2020 (Dassault Systemes, France), and mechanical tests were performed by using the open-source MD code LAMMPS (Sandia Labs, USA). All interactions of the constituent atoms were described by a polymer consistent force field (PCFF).[45]

4. RESULTS AND DISCUSSION

To identify the individual responses of the subconstituents comprising the molecular components, DGEBA and DETDA were divided into three categories (i.e., stiff part, flexible part, and side group) based on their structural characteristics, as shown in Figure 3a. The stiff part constitutes the backbone of the polymer network and is structurally rigid because of the presence of aromatic rings. The flexible part also constitutes the polymer network but can be easily bent or curled because of linear chains of hydrocarbons. The side group indicates the part that does not participate in the backbone of the polymer network. Using eq 3, we calculated the deformation energy associated with each part during uniaxial deformation.

Figure 3b shows the effective deformation energy density allocated to each molecular component. Because the epoxy microstructure modeled in this study consists only of the resin and hardener (no additives such as toughener or initiator), the sum of the deformation energy density of these two molecules is identical with the value of the entire system. Also, the results show that majority of the deformation energy of the system is associated with DGEBA because of its high molecular content. More specifically, when the results were classified according to covalent and noncovalent bonding, it was confirmed that DGEBA and DETDA both store most of the strain energy (77%) in covalent bonds. Such a substantial role of covalent bonding on the mechanical loading is in good agreement with findings previously reported in the literature.[43,46]

The effective contributions of the subconstituents of the molecules were also calculated, as shown in Figure 3c. For DGEBA, approximately half of the deformation energy was associated with the stiff part, and nearly all of the energy was involved in covalent bonding. In contrast, the flexible part exhibited considerable dependence on noncovalent intermolecular forces because of the hydroxyl groups created during the epoxide ring-opening reaction. For the side group, a relatively minor deformation was observed, and most of the deformation energy was expressed in the form of noncovalent interactions. A similar observation was made for the DETDA hardener. The stiff part and associated covalent energy are the primary factors to withstand external loads. The side group not belonging to the backbone of the epoxy networks was followed, and it contained a significant proportion of the effective deformation energy in the form of noncovalent energy.

The correlation between the cross-linking state and mechanical response at the molecular level was investigated. We measured the deformation energy of the free, dangling, and cross-linked chains, respectively. The energy values were divided by the weight fraction of molecules corresponding to each cross-linking state, and the results are shown in Figure 3d. At an elastic strain of 2%, the cumulative deformation energy was nearly constant, regardless of the cross-linking state. This was the same for both the DGEBA resin and DETDA hardener. However, at a strain of 4% or higher, the deformation energy was dominant in the cross-linked chain (~1.35-fold compared with the free and dangling chains). This trend continued up to a strain of 10%. This finding indicates that external deformation was converted to intramolecular stretching only in the case when both ends of the molecule are completely constrained. In the free or dangling components, the applied load was dispersed by an intermolecular slip rather than by internal deformations.

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Notably, an interesting result was also observed for the deformation energy distribution of the subconstituents of the molecules. In the cross-linked DGEBA chains, the deformation energy of the flexible part significantly increased, whereas the deformation energy value of the stiff part was nearly equal to that of the free and dangling chains. The linear structure of the flexible part absorbs a substantial portion of the applied internal energy because it is easily deformed by external loads.[47] This has a buffering effect that prevents load transfer to the stiff part. Therefore, the amount of deformation energy associated with the stiff part is maintained at a constant value, independent of the cross-linking state. Conversely, in the DETDA molecules, the stiff part was directly involved with covalent bonding during cross-linking reaction. Unlike the free chain that does not have restrictions on movement, the dangling chains of DETDA are forced to slide within the polymer network during the elongation. Moreover, the side groups in DETDA showed morphological resistance against the surrounding network; thus, the stiff part also participated in load transfer. Therefore, for the DETDA molecule, the influence of the stiff part increased proportionally as the cross-linking reaction proceeded.

For the DETDA hardeners reacted, the effective deformation energy was further classified with respect to the amine type (i.e., secondary or tertiary amines according to the number of cross-linked points), as shown in Figure 3e. The results showed that the deformation energy density of the dangling chains of DETDA hardeners was independent of the amine type. Because most of the applied energy is dissipated as intermolecular slip, additional bonding of amine groups does not make a difference in the load transfer properties. In contrast, the deformation energy of the cross-linked chains of DETDA hardeners increased in proportion to the tertiary amines content. DETDA containing at least one tertiary amine had a 1.9 times higher deformation energy than that composed of secondary amines at both ends, especially at strain values higher than 6%. DETDA composed of secondary amines exhibited a low load transfer capability comparable with that of free or dangling chains of DETDA hardeners. In the cross-linked epoxy network, DETDA with secondary amines only able to connect adjacent DGEBA molecules linearly. This may result in topological defects, such as loops or loose chains, which are detrimental to achieve effective resistance to external loads (see Figure S3). Therefore, DETDA becomes fully involved in the macromolecular network only when it has at least one tertiary amine, resulting in a high deformation energy during mechanical loading.

Figure 3f shows the effective deformation energy of each atom that constitutes the overall epoxy network structure divided by the number of corresponding atoms in the system. For the stiff part of the DGEBA resin, the majority of the deformation energy was associated with the linkages between the two aromatic rings (C1 and C2 atoms) and terminal group of the aromatic ring (C5 and O atoms). The value estimated for the aromatic rings (C3 and C4 atoms) was not significant because of its inherent rigidity. In the flexible part of the DGEBA resin, the C atoms in the backbone chain (C6 atoms) mainly contributed to the stretching of the covalent bonds. The hydroxyl groups (OH atoms) of the flexible constituents mostly formed noncovalent interactions. Because the interactions originated from intramolecular hydrogen bonding in the network, its contribution to the total deformation energy was comparable with that of the backbone carbon atoms. Similar results were obtained for the DETDA hardener. The terminal group of the aromatic ring (N and C8 atoms) showed a higher deformation energy than that of the aromatic ring (C7, C9, and C10 atoms) because of the differences in structural rigidity. Because almost all deformation energy appears as the stretching and angular bending of covalent bonds, it is self-evident that atoms located closer to the terminal group have a higher contribution to the total deformation energy (i.e., the largest deformation energy is exhibited by C7, followed by C9 and C10 atoms).

After completing a step-by-step analysis from the bulk network to the atomwise level, we identified the following physical attributes; the state of covalent bonding for each type of molecule constituting the epoxy network and its population density determine the mechanical properties of the overall material. Moreover, the intrinsic load-transfer properties of the material under mechanical loading obey the classical mechanics perspective, even with the classification of components at the atomic level. To better generalize the findings, epoxy microstructures with different chemical compositions were additionally prepared, and the deformation energy density during tensile loading was estimated. The compositional and structural details of the epoxy models are presented in Table 2. The models were constructed by changing the molecular ratio of DGEBA resin to DETDA hardener from 2:1 to 2:3. All models were prepared in the same manner as the original model previously described in section 3, and cross-linking simulations were run until 75% of the epoxide groups in DGEBA were reacted.

Table 2. Composition and Mechanical Properties of Epoxy Networks Modeled with Different Compositions

	no. of DGEBA	no. of DETDA	cross-linking ratio (%)	density (g/cm^3)	elastic modulus (GPa) 300 K	elastic modulus (GPa) 10 K
R100/H50 (reference model)	100	50	75.5	1.08	2.95	5.19
R100/H75	100	75	77.0	1.09	2.60	4.77
R100/H100	100	100	77.5	1.08	2.62	4.38
R100/H150	100	150	75.5	1.05	1.94	3.72

Figures 4a and 4b show the relative fraction of each molecule of DGEBA and DETDA, respectively, classified according to the cross-linking state. Because of the equal cross-linking ratio in all epoxy models, the fractions of free, dangling, and cross-linked chains in the DGEBA resin were nearly constant regardless of the hardener content. In contrast, the relative fraction of cross-linked DETDA significantly decreased as the amount of hardener increased. This caused the content of free and dangling compounds in the system gradually increased, which mechanically weakened the network structure. For the reacted DETDA, the molecular fraction of tertiary amines presented in the network decreased significantly, while the fraction of secondary amines increased accordingly. Changes in these molecular components also lead to differences in the mechanical properties of the bulk network, as shown in Figure 4c. The stress−strain responses were inversely proportional to the content of DETDA hardener because of the disappearance of cross-linked chains and tertiary

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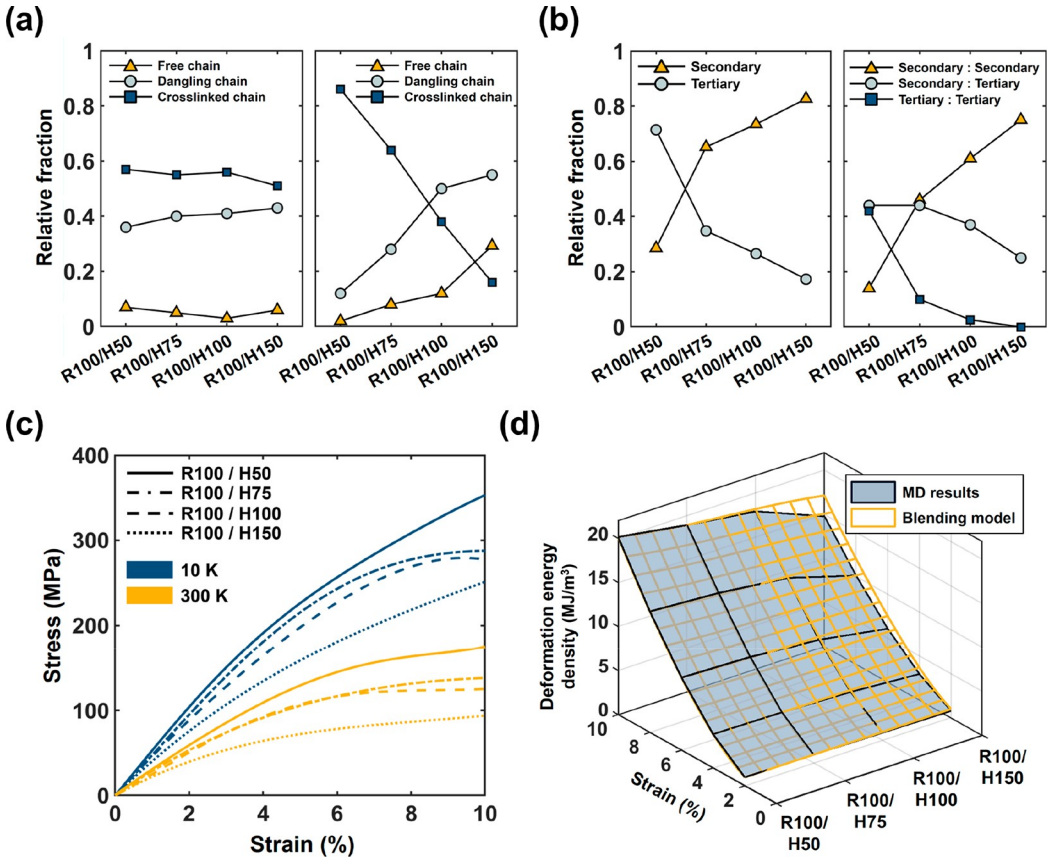


Figure 4. Topological and mechanical characterization of epoxy microstructures. (a) Relative fraction of free, dangling, and cross-linked chains in the epoxy network. Results for DGEBA resin (left) and DETDA hardener (right) are depicted separately. (b) Bonding state distribution of amine groups of the dangling (left) and cross-linked (right) chains of the DETDA hardener. (c) Stress−strain curves of epoxy microstructures obtained by MD simulation. Each result was fit with a fifth-order polynomial regression. (d) Comparison of the MD results and predictions obtained from the blending model for the increase in deformation energy density up to the maximum strain of 10% at a temperature of 10 K.

amines in DETDA. See also Table S2 and Figure S4 to assess the variance of the adjusted regression coefficients and raw stress−strain profiles for each model.

To directly predict the mechanical properties of the epoxy network based on the composition of the molecular-scale structure, we proposed a rule-of-mixture-based blending model formulated as follows

$$u_{\text{epoxy}} = \hat{u}_{\text{r}} w_{\text{r}} + \hat{u}_{\text{h}} w_{\text{h}} \tag{6-1}$$

$$w_{\text{r}} + w_{\text{h}} = 1 \tag{6-2}$$

where u and $\hat{u}$ indicate the deformation energy density and deformation energy density divided by the weight fraction, respectively, and w indicates the weight fraction of the components. The subscripts r and h denote the DGEBA resin and DETDA hardener, respectively. The formula for the DGEBA resin can be further specified according to the cross-linking state as follows:

$$\hat{u}_{\text{r}} w_{\text{r}} = \hat{u}_{\text{r}}^{\text{f}} w_{\text{r}}^{\text{f}} + \hat{u}_{\text{r}}^{\text{d}} w_{\text{r}}^{\text{d}} + \hat{u}_{\text{r}}^{\text{c}} w_{\text{r}}^{\text{c}} \tag{7-1}$$

$$w_{\text{r}}^{\text{f}} + w_{\text{r}}^{\text{d}} + w_{\text{r}}^{\text{c}} = w_{\text{r}} \tag{7-2}$$

where the superscripts f, d, and c indicate free, dangling, and cross-linked chain, respectively. Similarly, the formula for DETDA hardener can be further specified as follows:

$$\hat{u}_{\text{h}} w_{\text{h}} = \hat{u}_{\text{h}}^{\text{f}} w_{\text{h}}^{\text{f}} + \hat{u}_{\text{h}}^{\text{d}} w_{\text{h}}^{\text{d}} + \hat{u}_{\text{h}}^{\text{c}} w_{\text{h}}^{\text{c}} = \hat{u}_{\text{h}}^{\text{f}} w_{\text{h}}^{\text{f}} + (\hat{u}_{\text{h}}^{2} w_{\text{h}}^{2} + \hat{u}_{\text{h}}^{3} w_{\text{h}}^{3}) + (\hat{u}_{\text{h}}^{2:2} w_{\text{h}}^{2:2} + \hat{u}_{\text{h}}^{2:3} w_{\text{h}}^{2:3} + \hat{u}_{\text{h}}^{3:3} w_{\text{h}}^{3:3}) \tag{8-1}$$

$$w_{\text{h}}^{\text{f}} + (w_{\text{h}}^{2} + w_{\text{h}}^{3}) + (w_{\text{h}}^{2:2} + w_{\text{h}}^{2:3} + w_{\text{h}}^{3:3}) = w_{\text{h}} \tag{8-2}$$

where the superscripts "2" and "3" represent the dangling chains of DETDA with secondary and tertiary amines, respectively. The superscripts "2:2", "2:3", and "3:3" represent the cross-linked chains of DETDA with secondary:secondary, secondary:tertiary, and tertiary:tertiary amines, respectively. Details of the weight fractions of each molecular component can be found in Tables S3 and S4. By interpolating the weight fraction in each epoxy model and the effective deformation energy in the reference structure using a polynomial function, we continuously obtained a profile of the deformation energy of the system.

As shown in Figure 4d, the prediction results were in good agreement with those derived from the MD simulations for the R100/H75 and R100/H100 models (the errors are listed in Table S5). Reliable mechanical predictions of epoxy networks have been obtained, over the entire elastic−plasticity ranges, by using only the network topology and its distribution of deformation energy. However, in the R100/H150 model, where the content of DETDA hardener exceeds that of the DGEBA resin, the predicted values were overestimated compared with those obtained from the MD simulations. When the hardener was in excess of the resin, the components

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existed as individual segments rather than forming a single macromolecular network. This restricts intermolecular load transfer during the application of a mechanical load, which results in reduced effective work by the components. Therefore, it is not surprising that the values estimated on the premise of a macromolecular condensed state show a significant difference from the MD simulation results of the R100/H150 model. One should remind that the chemical composition of the R100/H150 model is far from that of ordinary epoxy materials in use. No epoxy system exists where the molecular ratio of the hardener exceeds that of the resin; hence, the practical investigation range should be set to the condition in which the molecular ratio of the resin to hardener is at most 1:1. Therefore, the findings of this study show that the mechanical properties can be accurately predicted over the full range under realistic compositional conditions.

5. CONCLUSION

In this study, we propose a novel computational method to interpret the mechanical behavior of highly cross-linked epoxy networks from a subcontinuum perspective. On the basis of the MD simulations and virial stress formula, the deformation energy contribution of the epoxy network under mechanical loading was classified according to the chemical bonding state. The results suggest that the DGEBA epoxy resin effectively withstands the load only in the case when both ends of the molecule are cross-linked. For the free and dangling segments where neither of the reactive sites at both ends form a covalent bond, they do not contribute significantly to the increase in the deformation energy density. Similarly, the mechanical response of the DETDA hardener was also sensitive to the amine type. The cross-linked DETDA molecule only contributed significantly to the increase in the deformation energy density when it contained one or more tertiary amines. Going one step further, we also performed individual analysis on the backbone atoms constituting the network and revealed how the applied load accumulates within the molecule.

The relationship between epoxy network topology and mechanical properties was analytically elucidated by the implementation of a rule-of-mixture-based blending model. The blending model was based on the weight fraction of molecules inserted, with the assumption that all members constituting the network have similar densities and are uniformly distributed. This simple approach enabled the prediction of the mechanical properties of epoxy thermosets with various compositions. By considering same molecules with different chemical bonds as independent components, a complete interpretation of the mechanical behavior of the epoxy network can be achieved. Therefore, the findings of this study clearly explain how the epoxy network structure established at the subcontinuum level determines its macroscopic properties.

This study provides credible guidelines for applying molecular-scale network topology to actual polymer design. The proposed blending model achieves a satisfactory level of prediction of mechanical properties for actual compositional conditions in the present state. In addition, there remains room for further development as a model for imperfectly interconnected networks caused by shortage or excess of hardener molecules. One can describe the load transfer efficiency of each molecule in the network as a function of its composition and topological structure to achieve this. It is also necessary to determine the characteristic dimensions at which the network begins to attain a saturation level of load transfer capacity. These challenges can be addressed straightforwardly in the near future because all the relevant data can be achieved by performing more MD simulations to analyze the topological changes of epoxy networks with different cross-linking ratios and compositional conditions. The extension of the model to more complex structures for polymer system problems, such as material anisotropy,[48,49] plasticity, fracture, and interfacial phase formation by nanoparticles,[50,51] is also underway.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.macromol.2c00593.

Considerations on the method; stress−strain profiles at different temperature conditions; topological defects formed by the DETDA; adjusted R^2 for polynomial regressions; stress−strain curve of epoxy models with different chemical compositions; number of molecules and the weight fraction of the epoxy components; number of molecules and the weight fraction of reacted DETDA; raw data of the results presented in Figure 4d (PDF)

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Notes

The authors declare no competing financial interest.

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